AMSE: all available evidence

Finding

Full AMSE improves spectral-power fidelity but sacrifices forecast accuracy. Lower AMSE weights retain more accuracy, yet no candidate in the completed joint campaign meets all three aggregate requirements plus the safety guard.

Coverage

September 19 full-AMSE/MSE pair; September 24 lambda=0.1 blend; October 4 trajectory re-evaluation, 500-update pilot, stopped/differentiable feedback, and three objective ablations. All 49 evaluated records across the four October phases are included, with repeated controls retained separately.

What is still pending

All six 2,000-update confirmation training runs finished. Seven evaluation tasks 15149394_0–6 and report 15149395 remain pending at the live queue check on October 7. Confirmation covers three seeds and 64 separate seasonal 2023 starts. No confirmation scores are available.

Related MGDA experiment

MGDA combines GraphCast and spectral-power gradients. Calibration completed; both 100-update smoke runs failed bitwise resume equality of residual parameters. The 2,000-update production comparison did not launch. These are engineering results, not evidence of forecast skill.

Reading this report

Positive reduction means better than Original GraphCast. Exact GraphCast loss is the headline metric; physical RMSE and relative spectral-power error are separate diagnostics. The companion complete PDF appends the existing 49-page field-level spectral report.